FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – III
PAPER –1
TEST DATE: 25-12-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

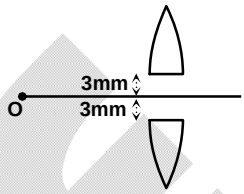
Physics

PART – I

Section – A (Maximum Marks: 24)

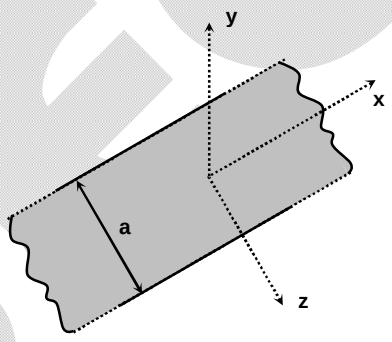
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

1. A point object is placed at a distance of 0.4 meters from a convex lens of focal length 0.2 m. The lens is cut into two equal parts each of which is displaced vertically by 3 mm away from the optic axis as shown in figure. Mark the correct statements.



(A) Image is formed at a distance of 0.4 meter from the lens
(B) Number of images formed are 2
(C) Separation between images is 12 mm
(D) Separation between images is 6 mm

2. A thin metallic plate of uniform mass density is kept along x-axis symmetrically. The length of plate along x-axis is very large as compared to its width 'a' along z-axis. It is given that 'σ' is mass per unit area. Then choose the correct option(s)



(A) The gravitational field at $\left(0,\frac{\sqrt{3}a}{2},0\right)$ is $\frac{2G\sigma\pi}{3}$.

(B) The gravitational field at $\left(0,\frac{a}{2},0\right)$ is $G\sigma\pi$.

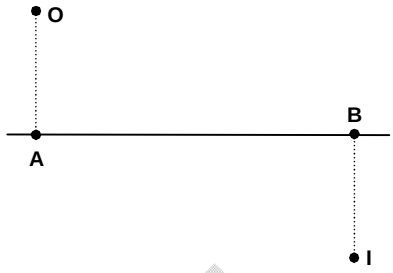
(C) The ratio of gravitational field at $\left(0,\frac{a}{2\sqrt{3}},0\right)$ and $\left(0,\frac{\sqrt{3}a}{2},0\right)$ is 2.

(D) The ratio of gravitational field at $\left(0,\frac{a}{2\sqrt{3}},0\right)$ and $\left(0,\frac{a}{2},0\right)$ is $\frac{4}{3}$.

3. Choose the correct option(s).
(A) If Q value of an exothermic nuclear reaction is +15.76 MeV and the kinetic energy of products of reaction is 18.24 MeV then kinetic energy of incident particles is 2.48 MeV.
(B) In a nuclear reaction, during α-decay from a nucleus at rest, energy released is shared between alpha particle and daughter nucleus is in the form of kinetic energy and its value for α-particle has more than that of daughter nucleus.
(C) In a decay process of a radioactive element, the activity of the sample at any instant is directly proportional to the number of undecayed nuclei present in the sample at that time.
(D) Photo electric effect supports quantum nature of light because there is a minimum frequency of light below which no photoelectrons are emitted.

4. In a resonance tube experiment, a closed organ pipe of length 125 cm resonates when tuned with a tuning fork of frequency 340 Hz. If water is poured in the pipe then (Given v_{air} = 340 m/s)
(A) Minimum length of water column to have the resonance is 50 cm
(B) The distance between two successive nodes is 50 cm
(C) Maximum length of water column to create the resonance in 100 cm
(D) Maximum length of water column to create the resonance in 105 cm

5. A luminous point object is placed at O, whose image is formed at I as shown in figure. Line AB is the optical axis. Which of the following statement is/are correct?



(A) If a lens is used to obtain the image, then it must be a converging lens and its optical center will be the intersection point of line AB and OI.
(B) If a lens is used to obtain the image, then it must be a diverging lens and its optical center will be the intersection point of line AB and OI.
(C) If a mirror is used to obtain the image then the mirror must be concave and object and image subtend equal angles at the pole of the mirror.
(D) I is a real Image.

6. Point A (0, 0.1 cm) and B (12 cm, 0.5 cm) are object image pair (one of the point acts as object and the other point as image) x-axis is the principal axis of the mirror. Then this object image pair is:
(A) Due to a convex mirror of focal length 2.5 cm
(B) Due to a concave mirror having its pole at (2cm, 0)
(C) Real virtual pair
(D) Data is insufficient for (A) and (B)

Section – A (Maximum Marks: 12)

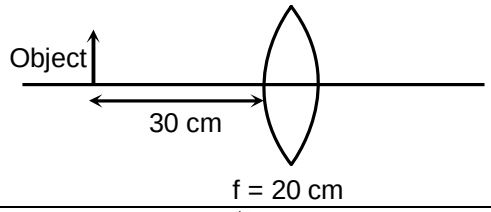
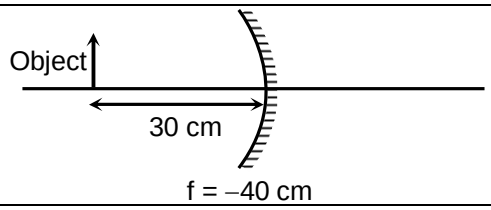
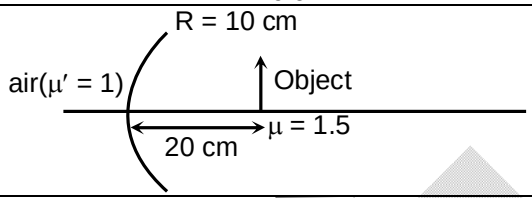
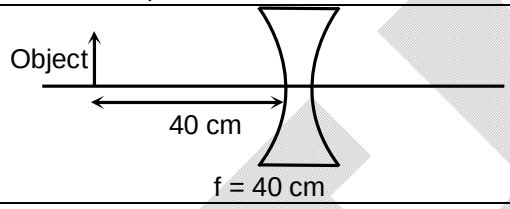
*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. List –I describes physical quantities and List-II gives their dimensions.

List –I		List -II	
(I)	Magnetic flux	(P)	$ML T^{-3} A^{-1}$
(II)	Electrical conductivity	(Q)	$M^{-1} L^{-3} T^{3} A^{2}$
(III)	Electric field	(R)	$M^{-1} L^{-2} T^{4} A^{2}$
(IV)	Capacitance	(S)	$ML^{2} T^{-2} A^{-1}$
		(T)	$M^{2} L^{2} T^{-2} A^{-1}$

Which one of the following options is correct ?
(A) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ P
(B) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ P, IV $\rightarrow$ P
(C) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ P
(D) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ R

8. List –I shows the object and optical device, List-II shows nature of the image formed for paraxial approximation. Match the List-I with nature of image in List-II.

List –I		List -II	
(I)		(P)	Erect image
(II)		(Q)	Inverted image
(III)		(R)	Magnified image
(IV)		(S)	Diminished image
		(T)	Real image

Which one of the following options is correct ?

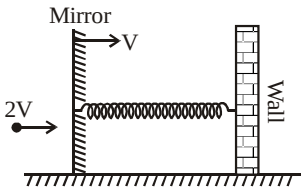
(A) I → Q,R,T II → P,S III → P,R IV → Q,R,T

(B) I → P,S II → Q,R,T III → P,S IV → P,R

(C) I → P,S II → P,R III → Q,R,T IV → P,S

(D) I → Q,R,T II → P,R III → P,R IV → P,S

9. A plane mirror is tied to the free end of an ideal spring. The other end of the spring is attached to a wall. The spring with mirror is held vertically to the floor, can slide along it smoothly. When the spring is at its natural length, the mirror is found to be moving at a speed of V with respect to ground frame. An object is moving towards the mirror with speed 2V with respect to ground frame. Then, Match the following :

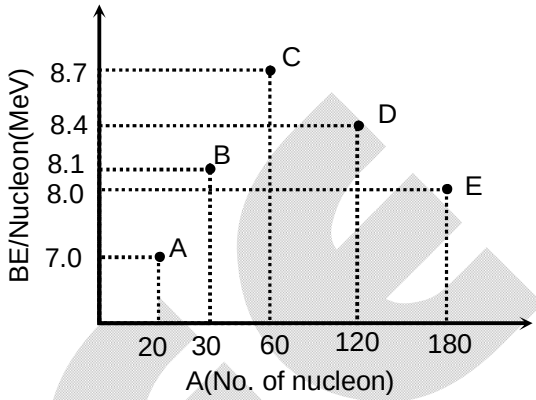


List –I		List –II	
(I)	Speed of image with respect to ground frame when spring is at natural length	(P)	V
(II)	Speed of image with respect to mirror when spring is at natural length	(Q)	O
(III)	Speed of image with respect to object when spring is at natural length	(R)	2V
(IV)	Speed of image with respect to ground frame when spring is at maximum compressed state	(S)	3V
		(T)	4V

Which one of the following options is correct ?

(A) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ S
(B) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P
(C) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ R
(D) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ P

10. Consider the position of different nuclei on the binding energy per nucleon graph as shown :



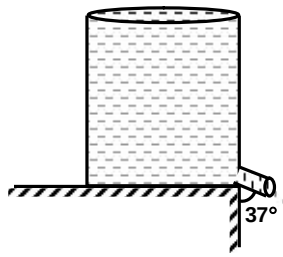
List –I (Reactions)		List –II [Q-values(MeV)]	
(I)	2B $\rightarrow$ C	(P)	Zero
(II)	2B + C $\rightarrow$ D	(Q)	102
(III)	E $\rightarrow$ C + D	(R)	Non-zero
(IV)	3A $\rightarrow$ C	(S)	36
		(T)	90

Which one of the following options is correct ?
(A) I $\rightarrow$ P,S II $\rightarrow$ P III $\rightarrow$ R,T IV $\rightarrow$ Q,R
(B) I $\rightarrow$ R,S II $\rightarrow$ P III $\rightarrow$ P,T IV $\rightarrow$ Q,R
(C) I $\rightarrow$ S,T II $\rightarrow$ P,R III $\rightarrow$ R,T IV $\rightarrow$ Q,R
(D) I $\rightarrow$ R,S II $\rightarrow$ P III $\rightarrow$ R,T IV $\rightarrow$ Q,R

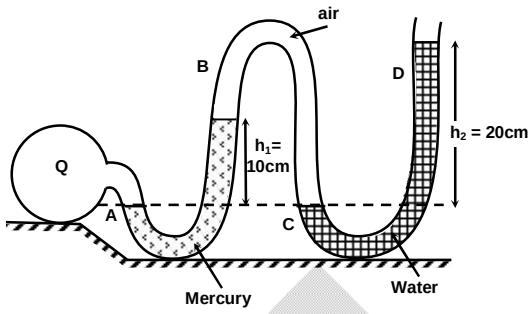
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

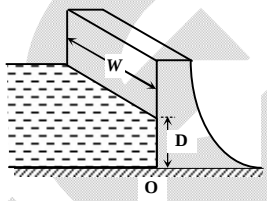
11. A cylindrical tank of cross-sectional area A filled with water upto certain height is placed on a rough horizontal surface having coefficient of friction $\mu = 0.5$. Water comes out of the tank through a small tube of cross-sectional area 'a' inclined at an angle $\theta = 37°$ with the wall of the tank as shown in the figure. Assuming Bernoulli's theorem to be applicable here, find the minimum ratio of a/A so, that the frictional force between tank and horizontal surface is limiting. Neglect the mass of the tank.



12. A bulb Q is filled with air. A tube is connected to the bulb filled with mercury and water. A column of air is trapped between mercury and water as shown in the figure. The heights of mercury and water are shown in the figure. Find the pressure inside the bulb Q(in mega N/m^2) assuming one end of tube opens to open atmosphere. Assume density of air to be small. (Take g = 10 m/s^2 and atmospheric pressure 10^5 N/m^2)



13. Water stands upto depth D, behind the vertical face of dam of width W. Given that it exerts a resultant horizontal force = 3000 kN and torque about O tending to overturn is 10000 kN-m. Find the width (in m). (Taking density of water = 1000 kg/m^3, g = 10 m/s^2)



14. A solid cylindrical piece of wood of length L = 0.5 m, cross-sectional area A = 1×10^{-2} m^2 and density ρ = 600 kg/m^3 is floating with its axis vertical in a liquid of density ρ_0 = 1200 kg/m^3. The wood is pressed down slowly into the liquid from its equilibrium position. Find the work done (in J) by the force pressing it until it is just immersed completely.

15. The resistance of a metal is given by R = V/I, where V is potential difference and I is the current. In a circuit the potential difference across resistance is V = (12$\pm$0.4)V and current in resistance I = (2$\pm$0.2)A. What is the maximum percentage error in measurement of resistance?

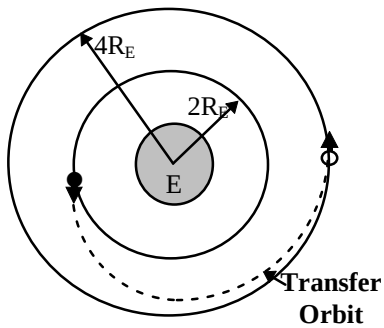
16. In a screw gauge, the graduation in the circular scale consists of 100 equal divisions and 10 complete rotation of the cap is equivalent to 5 mm on the main scale. While measuring diameter of a wire with screw gauge, a particular observation gives initial circular cap reading as 57 and after completing 3 complete rotation the final circular cap reading as 63. It is observed that while proceeding from initial reading to final reading the cap reading increases. Calculate the diameter of the wire (in mm) from the above data.

17. In the measurement of the focal length of a convex lens using the u-v method the following values were obtained:

u	35 cm	40 cm	50 cm	60 cm
v	46.2 cm	40.5 cm	33 cm	30.5 cm

Find the value of focal length by taking the average of the measurements. If the errors in u and v are 0.4 cm respectively, estimate the error in the measurement of focal length in a typical measurement (in cm).

18. An artificial satellite of mass m orbiting the earth in a circular orbit of radius twice the earth's radius R. It is to be taken in another circular orbit of radius 4R. The transfer is accomplished through an elliptical orbit as shown in the figure. The energy needed to complete the job is calculated to be equal to $\frac{1}{k}\left(\frac{GMm}{R}\right)$, where M is the mass of the earth. Find the value of k.



Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

19. 500 ml aqueous solution is made by dissolving a certain amount of Na_2CO_3 and NaOH mixture. 100 ml of this solution require 40 ml, 0.1 M HCl solution for the colour change when phenolphthalein is used as indicator, while 60 ml 0.1 N HCl require for the colour change when methyl orange is used as indicator in a separate titration. The correct option(s) is/are
(A) Weight % of NaOH in the mixture is 27.4.
(B) Weight of NaOH in the mixture is 0.08 gm.
(C) Weight of Na_2CO_3 in the mixture is 1.06 gm.
(D) Total weight of mixture is 0.292 gm.

20. The possible product(s) obtained by the hydrolysis (partial or complete) of XeF_6 is/are
(A) XeO_3
(B) $XeOF_4$
(C) XeO_3F_2
(D) XeO_2F_2

21. The INCORRECT statement(s) among the following is/are
(A) Multilayer adsorption of any gas on a solid surface is physisorption.
(B) The plot of log(x/m) vs. log(P) in the Freundlich adsorption isotherm is straight line with negative slope.
(C) High pressure and high temperature is favourable condition for physisorption.
(D) The magnitude for heat of adsorption of chemisorption is lower than physisorption.

22. Which of the following is correct?
(A) van Arkel method is used for the purification of Ti.
(B) Poling is used for purification of Cu.
(C) Mond's process is used for the purification of Au.
(D) Cupellation is used for the purification of Ag.

23. A gaseous mixture of 1 : 2 molar ratio of CH_4 and He diffuses two times slower than a gas 'X' under identical condition. Which of the following statement(s) is/are true about the gas 'X"?
(A) The gas "X' is O_2.
(B) The gas 'X' diffuse $\sqrt{2}$ times faster than He gas under identical condition.
(C) The gas 'X' is lightest known molecule.
(D) The gas 'X' diffuses 4 times slower than H_2 gas under identical condition.

24. A compound has two isomeric liquid 'A' and 'B' with pure vapour pressure $P_A^o = 100$ torr and $P_B^o = 80$ torr at 298 K. The binary solution of liquid A and B is in equilibrium with its vapour. The molar ratio of A and B in its liquid form is found to be 1 : 2 at room temperature. Which of the following options are correct?
(A) Mass ratio of A and B in vapour phase is 5 : 8.
(B) Mole ratio of A and B in vapour phase is 1 : 2.
(C) Mole ratio of A and B in vapour phase is 5 : 8.
(D) Total vapour pressure of the solution is $\frac{260}{3}$ torr .

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. List-I contains processes List-II contains metal or their compound used as catalyst

List – I		List – II	
(I)	Ziglar-Natta catalyst	(P)	V_2O_5
(II)	Ostwald process	(Q)	$TiCl_4$
(III)	Contact process	(R)	Fe
(IV)	Haber-Bosch process	(S)	Pt
		(T)	$Al(C_2H_5)_3$

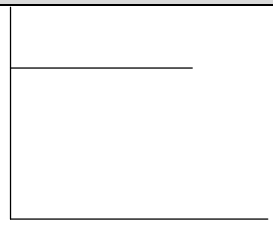
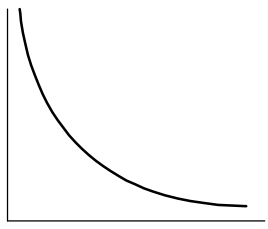
(A) I→Q; II→P; III→T; IV→R
(B) I→T; II→Q; III→R; IV→S
(C) I→Q,T; II→R; III→P; IV→S
(D) I→Q,T; II→S; III→P; IV→R

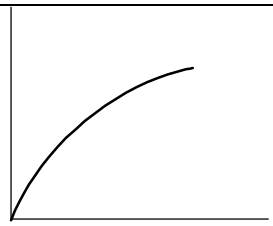
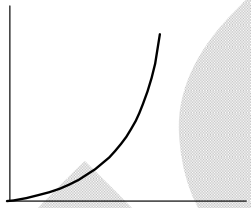
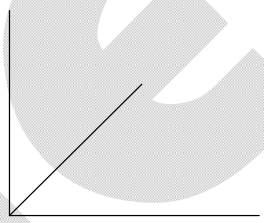
26. Match the List-I with List-II

List – I		List – II	
(I)	The process is mainly applicable for sulphide ore	(P)	Froth floatation
(II)	The process is mainly applicable for carbonates	(Q)	Calcination
(III)	Physical method of separation	(R)	Roasting
(IV)	Chemical method of separation	(S)	Leaching
		(T)	Concentration of the ore

(A) I→S; II→P; III→P,S; IV→Q
(B) I→P, R; II→Q; III→P; IV→S
(C) I→Q,R; II→P; III→T; IV→P
(D) I→R; II→T; III→Q; IV→S, R

27. Match the correct graph, if the process is taken place at constant temperature for the fixed amount of ideal gas

List – I		List – II	
(I)	P vs. $\frac{1}{V^2}$	(P)	
(II)	P vs. $\frac{1}{V}$	(Q)	

(III)	$\frac{1}{P^2}$ vs. V	(R)	
(IV)	PV vs. V	(S)	
		(T)	

(A) I→R; II→S; III→Q; IV→P
(B) I→S; II→T; III→R; IV→P
(C) I→S; II→Q; III→P; IV→R
(D) I→S; II→P; III→R; IV→R

28. List-I contains metal complexes and List-II contains their properties

List – I		List – II	
(I)	$\left[Cr(H_2O)_6\right]^{3+}$	(P)	Central atom is sp^3d^2 hybridisation
(II)	$\left[Co(CN)_6\right]^{3-}$	(Q)	Central atom is d^2sp^3 hybridized
(III)	$\left[CuF_6\right]^{4-}$	(R)	t_{2g} orbital contains 6 electron
(IV)	$\left[MnF_6\right]^{3-}$	(S)	Unsymmetrical e_g orbitation electronic configuration
		(T)	3-unpaired electron present

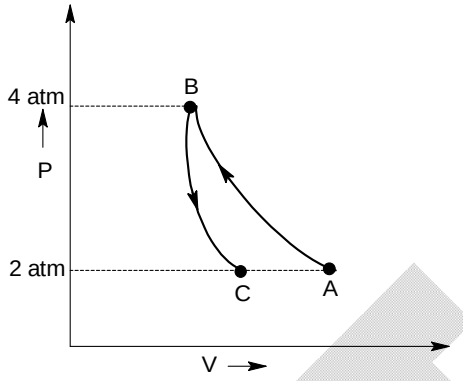
[Given atomic number of Cr = 24, Co = 27, Ni = 28, Mn = 25]
(A) I→Q,T; II→Q, R; III→Q,R; IV→P, S
(B) I→P,T; II→Q, R; III→R, S; IV→P,S
(C) I→Q,S, T; II→Q,R; III→P, R, S; IV→P, T
(D) I→Q,T; II→Q, R; III→P,S; IV→P, S

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. The EMF of the cell at 298 K
$Ag \mid AgBr(s), KBr(1\ M) \parallel AgNO_3(0.1\ M) / Ag(s)$ is 0.591 V. If in aq. solution $AgNO_3$ dissociate 80% and KBr dissociate 75%, the solubility of AgBr found to be $x \times 10^{-12}$ mol/litre. The value of x is

30. One mole of an ideal gas $\left(C_{p,m} = \frac{5}{2}R\right)$ at 500 K follows path $A \rightarrow B$ (Reversible isothermal) and then $B \rightarrow C$ (Reversible adiabatic) as shown in graph. What will be the volume of gas (in litre) at point C?
[Given $R = 0.082$, $2^{3/5} = 1.52$]

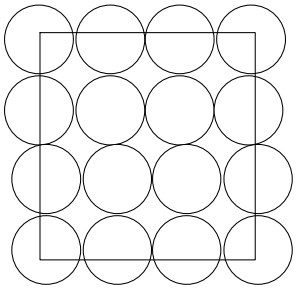


31. AgX crystallizes in rock salt like structure in which X^- occupy FCC lattice site. The density of AgX crystal found to be $\frac{40}{6.022}$ g / cm^3 and the edge length of the unit cell is 600 pm. What will be the mass of 1 mole unit cell if all the atoms along one body diagonal of each unit cell have been removed. [Given : Atomic weight of = 108 g/mol, $N_A = 6.022 \times 10^{23}$]

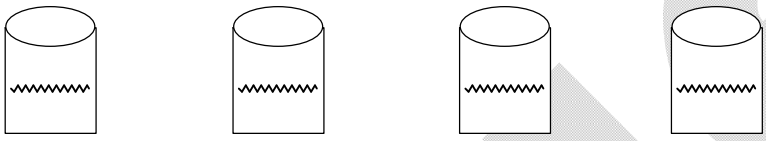
32. An excess amount of $CaCO_3$ is treated with 600 ml, 0.1 N HCl solution. The obtained $CaCl_2$ is used for the preparation of plaster of paris. The maximum amount of plaster of paris (in gram) that could be obtained is? [At. wt. of Ca = 40, S = 32, O = 16, C = 12, H = 1]

33. An aqueous solution of 0.2 M CH_3COOH is placed between parallel electrodes separated 1 cm apart with cross sectional area 1 cm^2. The resistance of solution is found to be $50\,\Omega$. What will be the osmotic pressure of same acetic acid solution at 300 K in atm?
[Given $\lambda^o_M\ CH_3COO^- = 50$ S cm^2 mol^{-1}, $\lambda^o_M\ H^+ = 350$ S cm^2 mol^{-1}, $R = \frac{1}{12}$ lit atm K^{-1} mol^{-1}

34. The % packing efficiency of two dimensional square close packed unit cell shown below is $(\pi = 3.14)$



35. All four identical beakers containing $H_2O(\ell)$ (shown below) is heated separately to reach temperature 90^oC, for this purpose w, x, y, and z Joule energy is required. If the magnitude of thermal energy required is w < x < y < z then what is the value of $\frac{x}{y}$. (Assuming there is no loss of $H_2O(\ell)$ during heating)



200 g 40^oC (A)

100 g 20^oC (B)

150 g 50^oC (C)

300 g 30^oC (D)

36. 45 gram of anhydrous oxalic acid is present in 500 ml solution. If 50 ml of this solution is shaken with 0.2 gram of wood charcoal, the concentration of oxalic acid become 0.6 molar. The amount of (in gm) oxalic acid adsorbed per gram of charcoal is [M.Wt. of C = 12, O = 16, H = 1). [Assuming volume of solution remains same after adsorption]

Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

37. Starting at (–3, –3) a bug moves in the co-ordinate plane via a sequence of steps, each of length one, each step is left, right, up or down all four being equally likely, if P_n is the probability that object reaches (–1, –1) first time in 'n' steps, then which of the following is/are CORRECT?

(A) $P_4 = \frac{3}{128}$ (B) $P_5 + P_6 = \frac{3}{128}$

(C) $P_4 + P_6 = \frac{3}{64}$ (D) $P_4 + P_5 = \frac{3}{64}$

38. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 4$, $|\vec{b}| = 3$ & $\vec{a}\cdot\vec{b} = 6\sqrt{3}$ and let $\vec{P}_1$ and $\vec{P}_2$ be two vectors such that $\vec{P}_1 = (\vec{a}\cdot\hat{b})\hat{b} - |\vec{a}|(\hat{a}\cdot\hat{b})^2\,\hat{a}$, $\vec{P}_2 = \vec{a} - (\vec{a}\cdot\hat{b})\hat{b} + \vec{P}_1$ and vector $\vec{P}_3$ is parallel to vector $\vec{a}\times\vec{b}$ such that $|\vec{P}_3| = 6$, then which of the following is/are CORRECT?

(A) $|\vec{P}_1| + |\vec{P}_2| = 1 + \sqrt{3}$ (B) $|\vec{P}_3 \cdot (\vec{P}_1 \times \vec{P}_2)| = 18\sqrt{3}$

(C) $|\vec{P}_1| + |\vec{P}_2| = 4 + \sqrt{3}$ (D) $|\vec{P}_3 \cdot (\vec{P}_1 \times \vec{P}_2)| = 6\sqrt{3}$

39. If $\sum_{r=0}^{1000} \frac{\left({}^{1000}C_r\right)}{\left({}^{2019}C_r\right)} = \frac{m}{n}$ (where m and n are relatively prime positive integers) and

$N = \left[\frac{m}{n}\right] + \left[\frac{2m}{n}\right] + ... + \left[\frac{(n-1)m}{n}\right]$ (where [.] represents greatest integer function), then which of the following is/are CORRECT?

(A) m + n = 100 (B) N is divisible by 3

(C) N is divisible by 5 (D) m – n = 50

40. If all the complex number z = x + iy, x, y ∈ R, $i = \sqrt{-1}$, such that points of complex coordinates z, z^2, z^3, z^4 in this given order, are the vertices of a cyclic quadrilateral ($z \neq 0 + i0, 1 + i0$), then

(A) $\arg z \in \left(0, \frac{2\pi}{3}\right)$ (B) $\arg z \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$

(C) $\frac{1+z+z^2}{z}$ is purely real (D) $\arg z \in \left(\frac{4\pi}{3}, 2\pi\right)$

41. Let M_1, M_2 be two matrices of order 3×3, all the elements of both the matrices are real numbers and $(M_1)^2 + (M_2)^2 = O$ (where O is null matrix), then which of the following statements is/are CORRECT?

(A) $\det(M_1) + \det(M_2) = 0$

(B) There exist a non-null matrix M of order 3×3, such $(MM_1 + MM_2)$ is invertible matrix

(C) There exist infinitely many ordered pair of (a, b), a, b ∈ R, such that $\det((aM_1) + (bM_2)) = 0$

(D) There exist infinitely many matrix M of order 3×3 such that $M(M_1 + M_2)$ is non-invertible matrix

42. Let z_1, z_2, z_3 be three complex numbers of unit modulus such that $z_1 + z_2 + z_3 \neq 0$ and $z_1^2 + z_2^2 + z_3^2 = 0$, then which of the following is/are CORRECT?

(A) $\left|z_1^3 + z_2^3 + z_3^3\right| = 1$ (B) $\left|z_1^{2021} + z_2^{2021} + z_3^{2021}\right| = 2$

(C) z_1^2, z_2^2, z_3^2 are collinear (D) $\left|z_1^{2023} + z_2^{2023} + z_3^{2023}\right| = 0$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. For any finite set S, let n(S) denote the number of elements in S, suppose A and B are two subset of {1, 2, 3, 4, 5} that satisfy $n(A)\cdot n(B) = n(A \cap B)\cdot n(A \cup B)$ and P(E) is the probability of occurrence of event E, then (where n(A) is number of elements in the set A)

List-I		List-II	
(I)	P(n(A) > n(B))	(P)	$\frac{32}{211}$
(II)	$P(n(A) \leq n(B))$	(Q)	$\frac{211}{454}$
(III)	$P\left(\frac{(n(A) = n(B))}{(n(A) \geq n(B))}\right)$	(R)	$\frac{32}{454}$
(IV)	P(n(A) = n(B))	(S)	$\frac{243}{454}$
		(T)	$\frac{32}{243}$

The correct option is:

(A) (I) → (P); (II) → (S); (III) → (T); (IV) → (Q)

(B) (I) → (Q); (II) → (S); (III) → (T); (IV) → (R)

(C) (I) → (T); (II) → (R); (III) → (P); (IV) → (Q)

(D) (I) → (Q); (II) → (T); (III) → (P); (IV) → (R)

44. Consider the matrices $M = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}$ such that det(M) = 1, a, b, c ∈R, then

List-I		List-II	
(I)	There is only one matrix M	(P)	$a \in \{-1, 0, 1\}$, $b \in \{-2, -1, 0\}$, $c \in \{1, 2, 3\}$
(II)	There are two distinct matrices M	(Q)	$a \in \{-1, 0, 1\}$, $b \in \{-2, 0, 1\}$, $c \in \{-2, -1, 0\}$
(III)	There are more than 2, but finitely many matrices M	(R)	$a, b, c \in \{0, 1, 2\}$
(IV)	There are infinitely many matrices M	(S)	a, b, c all are negative integers
		(T)	$a, b, c \in Q$

The correct option is:

(A) (I) → (P); (II) → (T); (III) → (R); (IV) → (S)

(B) (I) → (S); (II) → (Q); (III) → (R); (IV) → (T)

(C) (I) → (P); (II) → (R); (III) → (T); (IV) → (S)

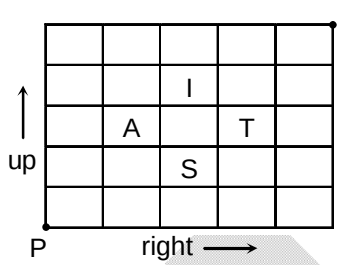
(D) (I) → (P); (II) → (Q); (III) → (R); (IV) → (T)

45. If $z_1, z_2, ..., z_8$ are the roots of the equation $(z^2 - 4z + 4)^4 = (9)^4$, such that arg z_1 < arg z_2 < ... < arg z_8. (where arg $z \in [0, 2\pi)$). Let z_k is a complex number satisfying the equation $|z_k - z_4| = |z_8 - z_k|$, and let
$L = |(z_k - z_2)(z_k - z_3)(z_k - z_4)(z_k - z_5)|$, then

List-I		List-II	
(I)	$z_k = 3 + i$	(P)	$L = 3^4$
(II)	$z_k = 5 + 3i$	(Q)	$L = \sqrt{\left(\frac{3\sqrt{2}-3}{3\sqrt{2}+3}\right)(3^8 \cdot 2^4 - 3^8)}$
(III)	$z_k = 2 + i0$	(R)	$L = \sqrt{\frac{1}{5}\cdot(3^8 - 2^8)}$
(IV)	$Z_k = 0 + i0$	(S)	$L = \sqrt{\left(\frac{3-\sqrt{2}}{3+\sqrt{2}}\right)(3^8 - 2^4)}$
		(T)	$L = 2^4$

The correct option is:
(A) (I) → (T); (II) → (P); (III) → (S); (IV) → (R)
(B) (I) → (S); (II) → (Q); (III) → (P); (IV) → (T)
(C) (I) → (T); (II) → (P); (III) → (R); (IV) → (S)
(D) (I) → (S); (II) → (Q); (III) → (P); (IV) → (R)

46. A square of 5 × 5 unit^2 area is shown is the diagram, contain 25 square of 1 × 1 unit^2 area. A person can move on this grid (on the sides of small squares) either right or up, then



List-I		List-II	
(I)	Number of such paths from point P to point Q which do not touch the boundary of squares (1 × 1 unit^2 area) contain alphabets A, I, T and S	(P)	73
(II)	Number of such paths from point P to point Q which do not crosses the diagonal PQ at any moment	(Q)	84
(III)	How many rectangles can be formed by joining small (1 × 1 unit^2 area) so that none of rectangles contain alphabets A, I, T and S	(R)	218
(IV)	Number of such paths from P to Q, which always pass through the boundaries of the square which contain either (A and I) or (T and S)	(S)	144
		(T)	4

The correct option is:
(A) (I) → (T); (II) → (P); (III) → (Q); (IV) → (R)
(B) (I) → (P); (II) → (Q); (III) → (T); (IV) → (S)
(C) (I) → (T); (II) → (Q); (III) → (P); (IV) → (R)
(D) (I) → (Q); (II) → (R); (III) → (P); (IV) → (T)

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. Each unit square of a 3 by 3 unit grid is to be cloured either green or blue for each square, either colour is equally likely. If the probability of obtaining a grid that does not have any 2 by 2 green square is $\frac{p}{q}$, where p, q $\in$ N and p, q are coprime, then the value of |p – q|

48. A person can take one step or two steps or three steps at a time. If "P" is the number of distinct ways he can cover 8 steps, then value of P is

49. Consider a polynomial $f(x) = a_1x^3 + a_2x^2 + a_3x + a_4$ such that $a_1, a_2, a_3, a_4 \in \{0, 1, 2, ... , 8, 9\}$, how many such polynomial of degree three or less than three such that $a_1 – a_2 + a_3 – a_4 + 9 = 0$

50. If $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$, a, b, c, d $\in$ R such that det(A) + Tr(A) = 2, then minimum value of $\det(A^2 + \det(A)A + Tr(A)\cdot I_2)$ is (where Tr(A) is trace of A and I_2 is identity matrix of order 2×2)

51. Let f(x) be a least degree unique polynomial with rational coefficients such that, when f(x) divided by $(x^2 + x + 1)$, the remainder is x + 2 and when f(x) divided by $(x^2 + 1)$, the remainder is 2x + 1, then find the sum of squares of coefficients of the polynomial f(x) is

52. In a tetrahedron OPQR, $\vec{p}, \vec{q}, \vec{r}$ are position vector of vertices P, Q and R respectively and O is origin. If OP = QR = α, OQ = PR = β, OR = PQ = γ and G_1 and G_2 are centroid of ΔPQR and ΔPOR respectively such that $OG_1 \perp QG_2$, then the value of $\frac{\alpha^2 + \gamma^2}{\beta^2}$ is

53. If $x^2 + px + q = 0$ has two distinct real roots and $(x^2 + 2x – 1)^2 + p(x^2 + 2x – 1) + q = 0$ has four distinct real roots $\alpha, \beta, \gamma, \delta$ such that $\alpha < \beta < \gamma < \delta$ and $\alpha, \beta, \gamma, \delta$ form an arithmetic sequence, then maximum values of (p – q) is

54. If $m = e^{i\frac{\pi}{6}}$ and $n = e^{i\frac{2\pi}{3}}$, where $i = \sqrt{-1}$ two complex numbers, then number of ordered pair of (p, q), p, q $\in$ integers and $1 \le p, q \le 100$ which satisfy the equation $(n)^q = i(m)^p$, is